

Statistics 170S first class presentation

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Class website <https://www.math.ucla.edu/~rebrova/170S.html>

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Probability classes (170A/E)

- Main object - **random variable**. For example, $X \sim N(0, 1)$.
- Main concept - **probability distribution**. For example,
 - Bernoulli
 - Gaussian
 - Chi-square
 - Gamma
 - Student
 - ...
- Ways to describe random variables and related properties:
 - density function, CDF (cumulative distribution function)
 - independence
 - expectation, variance
 - ...

Review of probability that we will need is in HW 1!

Probability classes (170A/E)

We have X_1, X_2, X_3, X_4, X_5 - independent standard normal random variables...

Questions typically asked:

1. What is the density function of X_2 ? Joint density of X_2 and X_5 ?
2. What is the expectation and variance of X_2 ?
3. Probability that both X_1 and X_5 are positive?

Check your answers on the last slide

Statistics class (now!)

We have X_1, X_2, X_3, X_4, X_5 - some observed data points (numbers)...

For example, we observed 0, 1, 1, 0, 0.

Questions typically asked:

1. Can this be the record of five consecutive flips of one coin (recording 0 for HEADS and 1 for TAILS)?
2. Ok then, is this coin fair?
3. How sure we are?

Statistics class (now!)

We have X_1, X_2, X_3, X_4, X_5 - some observed data points (numbers)...

For example, we observed 0, 1, 1, 0, 0.

Questions typically asked (in slightly more formal way):

1. Can all this data be from the same **Bernoulli distribution**?
2. Ok then, is $\mathbb{E}X = 0.5$?
3. How sure we are?

Another example

Suppose we are testing new cars. Let

1.1, 0.9, 1.4, 1.1, 1.05 *seconds*

be the times required for a glove compartment door to open completely.

- Can this data come from Bernoulli distribution? Probably not :(Why?
- How about standard normal distribution $N(0, 1)$? Not really :(Why?
- How about $N(1, \sigma^2)$ for some small σ ? Maybe...
What should be σ ? How confident we are in our "maybe"?

Why do we care?

- Lots of data:
we cannot look at every number and make sense out of it,
- Data sample (observed) is not all the data:
common assumption is that we do not have access to it all

Distribution helps us understand and predict unobserved data points. And to make data based conclusions better!

Data sample

Every problem will start with

$X_1, \dots, X_n$ - observed data sample.

We assume that $X_1, \dots, X_n$ is a result of **independent repetitions** of some experiment.

So, they can be modeled by **independent identically distributed** random variables.

Overview of the course-1

Chapter 6.

1. (6.1-6.3) What can we compute from data? Without guessing some specific distribution
2. (6.4-6.8) **Parameter estimation**: if we model data as $N(1, \sigma^2)$, *what should be σ ?*

Chapter 7.

1. **Confidence intervals**: *give a range* for the value of σ , “we are 90% sure that σ must be between 0.2 and 0.5.”

Overview of the course-2

MIDTERM.

Chapter 8.

1. **Hypothesis testing:** “I think σ must be 0.2 (this is my *hypothesis*). But my friend said σ has to be bigger than that ($\sigma > 0.2$ - *alternative hypothesis*). Should I be convinced or stick to my opinion?”

Chapter 9.

1. More special questions about data! :)

In Chapter 9

For example,

1. From the table of salaries of recent CS, Applied Math and Financial Math graduates, can we conclude that average salary is (in)dependent from major?
2. From the data about 5 different car models and 5 types of fuel, can we conclude that one fuel type is better than the others (irregardless of the car)? OR maybe there is just a great fit between certain car and fuel?
3. Looking at the digits generated by a random number generator, can we believe the machine is truly random?

FINAL.

The whole course in one sentence: we will learn to make good conclusions from data.

Homework 1 is posted on the website!

Questions?

1. What is the density of X_2 ?

$$f_{X_2}(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}$$

Joint density of X_2 and X_5 ?

$$f_{X_2, X_5}(t, s) = \frac{1}{2\pi} e^{-\frac{t^2 + s^2}{2}}$$

2. What is the expectation and variance of X_2 ?

$$\mathbb{E}X_2 = 0 \text{ and } \text{var } X_2 = 1$$

3. Probability that both X_1 and X_5 are positive?

$$1/4$$